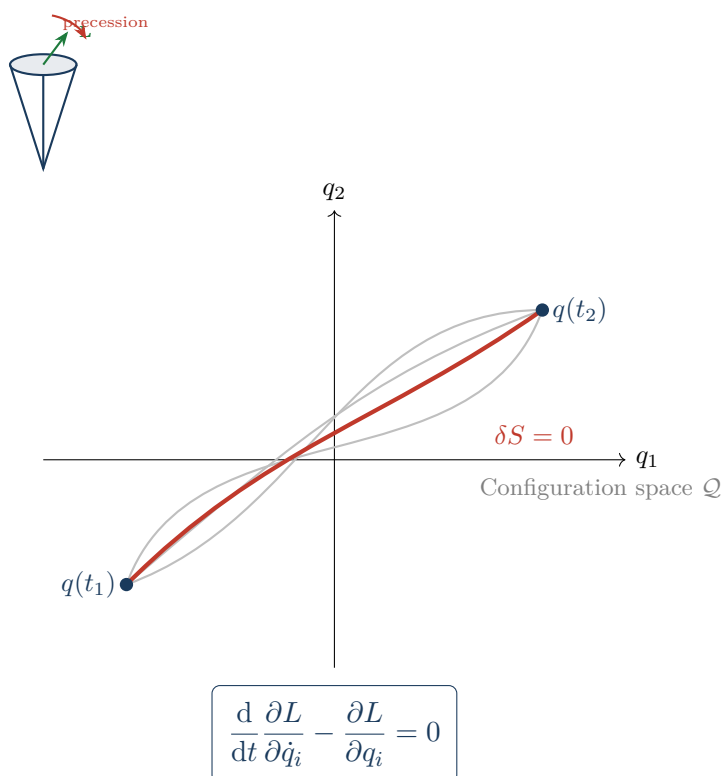


Advanced Mechanics

Comprehensive Notes



Topics covered:

Lagrangian Mechanics • Hamilton's Principle
Noether's Theorem • Conservation Laws
Small Oscillations • Rigid Body Dynamics
Euler Angles • Symmetric Tops • Hamiltonian Mechanics

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1 Foundations: Generalized Coordinates and Configuration Space

1.1 Degrees of Freedom and Generalized Coordinates

A mechanical system of N particles in 3D space has, in principle, $3N$ coordinates. **Constraints** reduce this number. The minimum number of independent coordinates needed to completely specify the configuration is the number of **degrees of freedom** d .

Definition 1.1 (Generalized Coordinates). *Any set of d independent quantities $q_1(t), q_2(t), \dots, q_d(t)$ that uniquely specify the configuration of the system are called **generalized coordinates**. They need not have dimensions of length — they can be angles, dimensionless ratios, or any other convenient parameterization.*

The collection of all possible configurations forms the **configuration space** $\mathcal{Q}$, which is a d -dimensional manifold (typically a subset of $\mathbb{R}^d$). A point $\mathbf{q}(t) = (q_1(t), \dots, q_d(t))$ in $\mathcal{Q}$ represents the instantaneous state of the system, and the system's time evolution traces a **path** through $\mathcal{Q}$.

State of the System

If all generalized coordinates q_i and generalized velocities $\dot{q}_i$ are specified at some time t_0 , the subsequent motion is completely determined (given the equations of motion). The pair $(q_i, \dot{q}_i)$ lives in the $2d$ -dimensional **state space** (or tangent bundle $T\mathcal{Q}$).

1.2 Holonomic Constraints

Definition 1.2 (Holonomic Constraint). *A constraint of the form*

$$f(q_1, q_2, \dots, q_d, t) = 0$$

*is called **holonomic**. It is a relationship among the coordinates (and possibly time) that can be used to eliminate one degree of freedom. Each holonomic constraint reduces d by one.*

Holonomic constraints **do no work** and do not need to appear explicitly in the Lagrangian formulation — they are built in by choosing the right generalized coordinates from the start.

Non-holonomic constraints involve velocities or inequalities (e.g., a ball rolling without slipping on a surface gives a constraint on $\dot{q}_i$ that cannot be integrated to a constraint on q_i alone). These are more subtle and require Lagrange multipliers or other methods.

Pendulum: From Cartesian to Generalized Coordinates

A simple pendulum (mass m , rod length ℓ) in the plane has two Cartesian coordinates (x, y) subject to the holonomic constraint $x^2 + y^2 = \ell^2$. This reduces the system to $d = 1$ degree of freedom. The natural generalized coordinate is the angle θ , with $x = \ell \sin \theta$ and $y = -\ell \cos \theta$.

2 The Lagrangian and Hamilton's Principle

2.1 The Lagrangian

The Lagrangian

For a mechanical system with kinetic energy T and potential energy V , the **Lagrangian** is:

$$L(q_i, \dot{q}_i, t) = T - V$$

The Lagrangian is a function of generalized coordinates, generalized velocities, and (possibly) time. It is a **scalar** — the same in every coordinate system.

This is a profound shift from Newton's formulation: instead of tracking vector forces, we work with a single scalar function.

2.2 The Action and Hamilton's Principle

Definition 2.1 (The Action Functional). *Given a path $q_i(t)$ from time t_1 to t_2 , the **action** is the integral of the Lagrangian along that path:*

$$S[q] = \int_{t_1}^{t_2} L(q_i(t), \dot{q}_i(t), t) dt$$

Note that S depends on the entire path $q(t)$, not just a single point — it is a **functional** (a function of functions).

Hamilton's Principle (Principle of Least Action)

Among all paths $q_i(t)$ connecting fixed endpoints $q_i(t_1)$ and $q_i(t_2)$, the physical path is the one that makes the action **stationary**:

$$\delta S = 0$$

That is, the first-order variation of S vanishes for the true path.

Why “Stationary” not “Minimum”?

Hamilton's principle requires $\delta S = 0$, which is a stationary point — analogous to $f'(x) = 0$ in calculus. The true path could be a minimum, maximum, or saddle point of S . For short time intervals, it is typically a minimum, but for longer paths it need not be. The historically common name “least action” is somewhat misleading.

2.3 Derivation of the Euler–Lagrange Equations

Consider a small perturbation $q_i(t) \rightarrow q_i(t) + \epsilon \eta_i(t)$ where $\eta_i(t_1) = \eta_i(t_2) = 0$ (the endpoints are fixed). The variation of the action is:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_i} \eta_i + \frac{\partial L}{\partial \dot{q}_i} \dot{\eta}_i \right) dt$$

Integrating the second term by parts (the boundary terms vanish since $\eta_i = 0$ at the endpoints):

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right) \eta_i dt = 0$$

Since $\eta_i(t)$ is arbitrary, the integrand must vanish identically:

The Euler–Lagrange Equations

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \quad i = 1, 2, \dots, d$$

These are d second-order ODEs that determine the motion of the system. They are equivalent to Newton’s second law but vastly more general.

2.4 Uniqueness of the Lagrangian

The Lagrangian is *not* unique. Two Lagrangians yield the same equations of motion if they differ by a **total time derivative** of an arbitrary function $F(q_i, t)$:

$$L'(q_i, \dot{q}_i, t) = L(q_i, \dot{q}_i, t) + \frac{dF(q_i, t)}{dt}$$

This is because the extra term integrates to a boundary contribution $F(q(t_2), t_2) - F(q(t_1), t_1)$ in the action, which does not depend on the path and therefore vanishes under variation.

Free Particle Lagrangian

For a free particle in 3D, $V = 0$ and $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. The Lagrangian $L = \frac{1}{2}m\dot{\mathbf{r}}^2$ is the same in every inertial frame (Galilean invariance). The E-L equations give $m\ddot{\mathbf{r}} = 0$: uniform motion.

Spherical Pendulum

A mass m is constrained to move on a sphere of radius ℓ . Using spherical coordinates (θ, ϕ) :

$$T = \frac{1}{2}m\ell^2(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta), \quad V = -mg\ell \cos \theta$$

$$L = \frac{1}{2}m\ell^2(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + mg\ell \cos \theta$$

The E-L equations are:

$$\theta : m\ell^2 \ddot{\theta} = m\ell^2 \dot{\phi}^2 \sin \theta \cos \theta - mg\ell \sin \theta$$

$$\phi : \frac{d}{dt}(m\ell^2 \dot{\phi} \sin^2 \theta) = 0$$

The second equation immediately gives conservation of the z -component of angular momentum: $L_z = m\ell^2 \dot{\phi} \sin^2 \theta = \text{const.}$

3 Symmetries and Conservation Laws

3.1 Cyclic Coordinates and Conserved Momenta

Definition 3.1 (Cyclic (Ignorable) Coordinate). *A generalized coordinate q_i is **cyclic** if the Lagrangian does not depend on it:*

$$\frac{\partial L}{\partial q_i} = 0$$

When q_i is cyclic, the E-L equation immediately gives:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0 \quad \implies \quad p_i \equiv \frac{\partial L}{\partial \dot{q}_i} = \text{const.}$$

Conjugate Momentum

The quantity

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

is the **generalized (canonical) momentum conjugate to q_i** . If q_i is cyclic, then p_i is conserved.

Note: the generalized momentum p_i is *not* always $m\dot{q}_i$. For example, for a charged particle in an electromagnetic field, p_i includes a contribution from the vector potential.

3.2 Noether's Theorem

The deep connection between symmetries and conservation laws is codified by one of the most beautiful results in physics:

Definition 3.2 (Symmetry). A **continuous symmetry** is a one-parameter family of transformations $q_i(t) \rightarrow Q_i(t, \epsilon)$ (with $Q_i(t, 0) = q_i(t)$) that leaves the equations of motion unchanged. Equivalently, the Lagrangian changes by at most a total time derivative.

Noether's Theorem

Every continuous symmetry of the Lagrangian corresponds to a conserved quantity.

If $q_i \rightarrow q_i + \epsilon K_i(q, t)$ is a symmetry (i.e., $\delta L = \epsilon \frac{dF}{dt}$ for some F), then the **Noether charge**:

$$I = \sum_i p_i K_i - F = \text{const.}$$

is conserved along solutions of the E-L equations.

3.3 The Great Conservation Laws from Noether's Theorem

Symmetry $\longleftrightarrow$ Conservation Law

Symmetry	Transformation	Conserved Quantity
Spatial translation	$\mathbf{r}_i \rightarrow \mathbf{r}_i + \epsilon \hat{\mathbf{n}}$	Linear momentum $\mathbf{P} \cdot \hat{\mathbf{n}}$
Time translation	$t \rightarrow t + \epsilon$	Energy E (Hamiltonian H)
Rotation about $\hat{\mathbf{n}}$	$\mathbf{r}_i \rightarrow \mathbf{r}_i + \epsilon \hat{\mathbf{n}} \times \mathbf{r}_i$	Angular momentum $\mathbf{L} \cdot \hat{\mathbf{n}}$

3.3.1 Conservation of Linear Momentum

For a closed system (no external forces), the Lagrangian is invariant under a uniform spatial translation $\mathbf{r}_i \rightarrow \mathbf{r}_i + \epsilon \hat{\mathbf{n}}$. By Noether's theorem:

$$\mathbf{P} = \sum_i m_i \dot{\mathbf{r}}_i = \text{const.}$$

3.3.2 Conservation of Angular Momentum

If the Lagrangian is invariant under rotations about an axis $\hat{\mathbf{n}}$, the infinitesimal transformation is $\delta \mathbf{r}_i = \epsilon \hat{\mathbf{n}} \times \mathbf{r}_i$, and:

$$\mathbf{L} \cdot \hat{\mathbf{n}} = \sum_i m_i (\mathbf{r}_i \times \dot{\mathbf{r}}_i) \cdot \hat{\mathbf{n}} = \text{const.}$$

For full rotational symmetry, the entire angular momentum vector $\mathbf{L}$ is conserved.

3.3.3 Conservation of Energy

If the Lagrangian has no explicit time dependence ($\partial L / \partial t = 0$), the transformation $q_i(t) \rightarrow q_i(t + \epsilon)$ is a symmetry. The conserved quantity is the **Hamiltonian**:

$$H = \sum_i p_i \dot{q}_i - L$$

When Does $H = E$?

The Hamiltonian equals the total energy $E = T + V$ when **both** of the following hold:

1. The kinetic energy is a **homogeneous quadratic** function of the $\dot{q}_i$ (no linear or constant terms in velocities).
2. The potential V does not depend on velocities.

If the coordinates are “natural” (the relation $\mathbf{r}_i = \mathbf{r}_i(q_1, \dots, q_d)$ has no explicit time dependence), then condition 1 is automatically satisfied.

When the coordinate transformation depends explicitly on time (e.g., rotating frames), T acquires linear terms in $\dot{q}_i$, and $H \neq E$.

H conserved $\nRightarrow H = E$

A system can have $\partial L / \partial t = 0$ (so H is conserved) but use time-dependent coordinate relations (so T is not purely quadratic in $\dot{q}$). In this case H is conserved but does not equal $T + V$.

3.4 Seven Conserved Quantities of a Closed System

Any closed system (no external forces or torques, no explicit time dependence) possesses **7 independent conserved quantities**:

1. 3 components of total linear momentum $\mathbf{P}$ (from translational symmetry)
2. 3 components of total angular momentum $\mathbf{L}$ (from rotational symmetry)
3. 1 total energy E (from time-translation symmetry)

3.5 Centre-of-Mass Motion and Reduced Lagrangian

For a two-body system with interaction $V(|\mathbf{r}_1 - \mathbf{r}_2|)$, define:

$$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}, \quad \mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$$

The Lagrangian separates:

$$L = \underbrace{\frac{1}{2} M \dot{\mathbf{R}}^2}_{\text{CM motion}} + \underbrace{\frac{1}{2} \mu \dot{\mathbf{r}}^2 - V(r)}_{\text{relative motion}}$$

where $M = m_1 + m_2$ is the total mass and $\mu = m_1 m_2 / M$ is the **reduced mass**. The CM moves freely, and the relative coordinate obeys a one-body problem.

4 The Euler–Lagrange Equations in Practice

4.1 Recipe for Setting Up a Lagrangian Problem

Lagrangian Problem-Solving Algorithm

1. **Identify degrees of freedom.** Count particles, subtract holonomic constraints.
2. **Choose generalized coordinates** $q_1, \dots, q_d$ that respect the constraints.
3. **Express T and V** in terms of q_i and $\dot{q}_i$. Write $\mathbf{r}_k = \mathbf{r}_k(q_1, \dots, q_d, t)$ for each particle, compute $\dot{\mathbf{r}}_k$, and form $T = \sum \frac{1}{2} m_k |\dot{\mathbf{r}}_k|^2$.
4. **Form $L = T - V$.**
5. **Identify cyclic coordinates** and write down conserved momenta immediately.
6. **Apply E-L equations** for each non-cyclic coordinate.
7. **Check:** Is energy conserved? Are there enough integrals of motion to solve the problem?

4.2 Lagrange Multipliers for Constraints

Sometimes it is useful to keep redundant coordinates and impose constraints explicitly. If we have a holonomic constraint $f(q_1, \dots, q_d, t) = 0$, the modified E-L equations become:

E-L Equations with Constraints

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \lambda \frac{\partial f}{\partial q_i}$$

where $\lambda(t)$ is a **Lagrange multiplier**. The right-hand side gives the **generalized constraint force**. This is useful when you *want* to find the constraint force (e.g., the tension in a string, or when the constraint breaks).

Bead on a Rotating Hoop

A bead slides freely on a circular hoop of radius r whose centre is at distance R from a vertical rotation axis. The hoop rotates at constant angular velocity ω . The single degree of freedom is the angle θ along the hoop:

$$x(t) = R \sin(\omega t) + r \sin(\omega t + \theta), \quad y(t) = R \cos(\omega t) + r \cos(\omega t + \theta)$$

After computing T and simplifying, the Lagrangian becomes:

$$L = \frac{1}{2} m \left[r^2 (\omega + \dot{\theta})^2 + 2 R r \omega (\omega + \dot{\theta}) \cos \theta \right]$$

The E-L equation gives, after linearization for small θ :

$$\ddot{\theta} + R \omega^2 \theta \approx 0$$

This is simple harmonic motion with frequency $\omega_0 = \omega \sqrt{R/r}$ (assuming small oscillations about $\theta = 0$). Note how neatly the Lagrangian approach handles the rotating frame.

5 Small Oscillations and Normal Modes

5.1 Linearization about Equilibrium

For a system with d degrees of freedom near a stable equilibrium $q_i = q_i^{(0)}$, define displacements $\eta_i = q_i - q_i^{(0)}$. Expand the Lagrangian to second order:

$$T \approx \frac{1}{2} \sum_{i,j} T_{ij} \dot{\eta}_i \dot{\eta}_j, \quad V \approx V_0 + \frac{1}{2} \sum_{i,j} V_{ij} \eta_i \eta_j$$

where $T_{ij} = \frac{\partial T}{\partial \dot{q}_i} \frac{\partial}{\partial \dot{q}_j} \big|_0$ is the **mass matrix** (or kinetic matrix) and $V_{ij} = \frac{\partial^2 V}{\partial q_i^2} \big|_0$ is the **stiffness matrix** (both evaluated at equilibrium).

The linearized equations of motion become:

$$\sum_j T_{ij} \ddot{\eta}_j + \sum_j V_{ij} \eta_j = 0 \quad \text{or} \quad \mathbf{T} \ddot{\boldsymbol{\eta}} + \mathbf{V} \boldsymbol{\eta} = 0$$

5.2 Normal Mode Analysis

Look for solutions of the form $\boldsymbol{\eta}(t) = \mathbf{a} e^{i\omega t}$. Substituting gives the **generalized eigenvalue problem**:

Normal Mode Eigenvalue Problem

$$\mathbf{V} \mathbf{a} = \omega^2 \mathbf{T} \mathbf{a}$$

The eigenvalues $\omega_1^2, \dots, \omega_d^2$ are the squared **normal mode frequencies**. The eigenvectors $\mathbf{a}^{(k)}$ describe the pattern of motion in each mode. The general solution is:

$$\boldsymbol{\eta}(t) = \sum_{k=1}^d C_k \mathbf{a}^{(k)} \cos(\omega_k t + \phi_k)$$

Key Properties of Normal Modes

1. There are exactly d normal mode frequencies for a system with d degrees of freedom.
2. All parts of the system oscillate at the *same* frequency in a given normal mode.
3. Normal modes are orthogonal with respect to $\mathbf{T}$: $\mathbf{a}^{(k)\top} \mathbf{T} \mathbf{a}^{(\ell)} = 0$ for $k \neq \ell$.
4. In the linear regime, energy does not transfer between normal modes.
5. Zero-frequency modes ($\omega = 0$) correspond to overall translations or rotations of the system.

Two Coupled Pendula

Two identical pendula (mass m , length ℓ) are connected by a spring (constant k) attached at their bobs. For small angles θ_1, θ_2 :

$$T = \frac{1}{2} m \ell^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2), \quad V = \frac{1}{2} m g \ell (\theta_1^2 + \theta_2^2) + \frac{1}{2} k \ell^2 (\theta_1 - \theta_2)^2$$

The mass and stiffness matrices are:

$$\mathbf{T} = m \ell^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} m g \ell + k \ell^2 & -k \ell^2 \\ -k \ell^2 & m g \ell + k \ell^2 \end{pmatrix}$$

The normal modes are:

Symmetric mode ($\theta_1 = \theta_2$): $\omega_1 = \sqrt{g/\ell}$ — the pendula swing together; the spring does nothing.

Antisymmetric mode ($\theta_1 = -\theta_2$): $\omega_2 = \sqrt{g/\ell + 2k/m}$ — the pendula swing oppositely; the spring is maximally stretched.

6 Rigid Body Mechanics

6.1 Degrees of Freedom of a Rigid Body

A rigid body in 3D has **6 degrees of freedom**: 3 for the position of the centre of mass and 3 for the orientation (rotations). The constraint of rigidity is $|\mathbf{r}_i - \mathbf{r}_j| = \text{const.}$ for all pairs.

6.2 The Inertia Tensor

The rotational kinetic energy of a rigid body rotating about a fixed point is:

$$T_{\text{rot}} = \frac{1}{2} \sum_{\alpha} m_{\alpha} |\boldsymbol{\omega} \times \mathbf{r}_{\alpha}|^2 = \frac{1}{2} \sum_{i,j} I_{ij} \omega_i \omega_j = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I} \boldsymbol{\omega}$$

Inertia Tensor

The **inertia tensor** is a 3×3 symmetric matrix with components:

$$I_{ij} = \sum_{\alpha} m_{\alpha} (|\mathbf{r}_{\alpha}|^2 \delta_{ij} - r_{\alpha,i} r_{\alpha,j})$$

For a continuous body: $I_{ij} = \int \rho(\mathbf{r}) (|\mathbf{r}|^2 \delta_{ij} - r_i r_j) dV$. The angular momentum is:

$$\mathbf{L} = \mathbf{I} \boldsymbol{\omega}$$

Important: $\mathbf{L}$ and $\boldsymbol{\omega}$ are generally *not* parallel.

6.3 Principal Axes and Principal Moments

Since $\mathbf{I}$ is real and symmetric, it can always be diagonalized by choosing the right coordinate axes.

Principal Axes

There exists an orthogonal set of axes (the **principal axes**) in which the inertia tensor is diagonal:

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

The diagonal elements I_1, I_2, I_3 are the **principal moments of inertia**. In this frame:

$$T_{\text{rot}} = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2), \quad L_i = I_i \omega_i$$

Classification of Rigid Bodies

Type	Condition	Examples
Spherical top	$I_1 = I_2 = I_3$	Uniform sphere, cube
Symmetric top	$I_1 = I_2 \neq I_3$	Cylinder, cone, football
Asymmetric top	$I_1 \neq I_2 \neq I_3$	General body

6.4 Transformation of the Inertia Tensor

Under a rotation described by matrix $\mathbf{R}$ (so that $\mathbf{r}' = \mathbf{R}\mathbf{r}$), the inertia tensor transforms as:

$$\mathbf{I}' = \mathbf{R} \mathbf{I} \mathbf{R}^T$$

This is the transformation law of a rank-2 tensor.

Parallel Axis Theorem (Tensor Form)

If $\mathbf{I}_{\text{CM}}$ is the inertia tensor about the centre of mass, then about a parallel set of axes displaced by $\mathbf{d}$:

$$I_{ij} = (I_{\text{CM}})_{ij} + M(|\mathbf{d}|^2 \delta_{ij} - d_i d_j)$$

6.5 Computing Inertia Tensors by Integration

For a body of uniform density ρ parameterized by coordinates (u, v, w) :

$$I_{ij} = \rho \iiint (\mathbf{r}^2 \delta_{ij} - r_i r_j) |J| du dv dw$$

where $|J|$ is the Jacobian determinant.

Inertia Tensor of a Uniform Ellipsoid

An ellipsoid with semi-axes a, b, c and uniform density $\rho = 3M/(4\pi abc)$ has principal moments (about its centre):

$$I_1 = \frac{1}{5}M(b^2 + c^2), \quad I_2 = \frac{1}{5}M(a^2 + c^2), \quad I_3 = \frac{1}{5}M(a^2 + b^2)$$

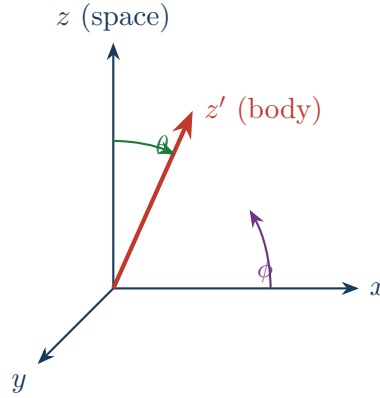
This can be derived by substituting $(x, y, z) = (a\tilde{x}, b\tilde{y}, c\tilde{z})$ to map the ellipsoid to a unit sphere, computing the integral over the sphere, and transforming back.

7 Euler Angles and Rigid Body Motion

7.1 Euler Angles

To describe the orientation of a rigid body, we use three **Euler angles** (ϕ, θ, ψ) , defined as successive rotations:

1. Rotate by ϕ about the **space-fixed** z -axis (precession).
2. Rotate by θ about the new x' -axis — the **line of nodes** (nutation).
3. Rotate by ψ about the new z'' -axis — the **body-fixed** symmetry axis (spin).



7.2 Angular Velocity in Terms of Euler Angles

The angular velocity expressed in the **body-fixed frame** is:

Body-Frame Angular Velocity Components

$$\omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi$$

$$\omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi$$

$$\omega_3 = \dot{\phi} \cos \theta + \dot{\psi}$$

For a **symmetric top** ($I_1 = I_2$), the rotational kinetic energy simplifies beautifully:

$$T_{\text{rot}} = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2$$

7.3 Euler's Equations of Motion

In the body-fixed frame, the angular momentum equation $\dot{\mathbf{L}} = \boldsymbol{\tau}$ becomes:

Euler's Equations

$$I_1 \dot{\omega}_1 - (I_2 - I_3) \omega_2 \omega_3 = \tau_1$$

$$I_2 \dot{\omega}_2 - (I_3 - I_1) \omega_3 \omega_1 = \tau_2$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2) \omega_1 \omega_2 = \tau_3$$

The extra terms $(I_j - I_k) \omega_j \omega_k$ arise because the body-fixed frame is itself rotating.

Why Body-Fixed Frame?

In the space-fixed frame, $\mathbf{I}$ changes with time as the body rotates, making the equations intractable. In the body-fixed frame, $\mathbf{I}$ is constant (diagonal if we choose principal axes), and the price is the extra “Euler terms” from the rotating frame.

7.4 Torque-Free Motion of a Symmetric Top

For a torque-free symmetric top ($I_1 = I_2$, $\tau = 0$), Euler's equations give:

$$\begin{aligned} I_1 \dot{\omega}_1 &= (I_1 - I_3) \omega_3 \omega_2 \\ I_1 \dot{\omega}_2 &= -(I_1 - I_3) \omega_3 \omega_1 \\ I_3 \dot{\omega}_3 &= 0 \end{aligned}$$

The third equation gives $\omega_3 = \text{const.}$ Define $\Omega = (I_1 - I_3) \omega_3 / I_1$; then $\dot{\omega}_1 = \Omega \omega_2$, $\dot{\omega}_2 = -\Omega \omega_1$, which means $\boldsymbol{\omega}_\perp = (\omega_1, \omega_2)$ rotates uniformly at rate Ω about the symmetry axis. This is **body-frame precession** (the angular velocity vector traces a cone around the body's symmetry axis).

8 The Symmetric Top with Gravity

8.1 Lagrangian of the Heavy Symmetric Top

A symmetric top (pivot fixed, $I_1 = I_2$) in a (constant) gravitational field has:

Heavy Symmetric Top Lagrangian

$$L = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 - M g \ell \cos \theta$$

where ℓ is the distance from the pivot to the centre of mass.

8.2 Conserved Quantities

Both ϕ and ψ are cyclic, giving two conserved momenta:

$$\begin{aligned} p_\phi &= I_1 \dot{\phi} \sin^2 \theta + I_3 (\dot{\psi} + \dot{\phi} \cos \theta) \cos \theta = L_z \quad (\text{space-}z \text{ angular momentum}) \\ p_\psi &= I_3 (\dot{\psi} + \dot{\phi} \cos \theta) = L_3 \quad (\text{body-}z \text{ angular momentum, i.e., spin}) \end{aligned}$$

Additionally, since $\partial L / \partial t = 0$ and T is quadratic in velocities, the total energy:

$$E = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 + M g \ell \cos \theta$$

is conserved. With **three integrals of motion** (E , p_ϕ , p_ψ) for three degrees of freedom, the system is **completely integrable**.

8.3 Effective Potential for Nutation

Using p_ϕ and p_ψ to eliminate $\dot{\phi}$ and $\dot{\psi}$, the energy becomes:

$$E = \frac{1}{2} I_1 \dot{\theta}^2 + V_{\text{eff}}(\theta)$$

where:

Effective Potential for the Heavy Top

$$V_{\text{eff}}(\theta) = \frac{(p_\phi - p_\psi \cos \theta)^2}{2 I_1 \sin^2 \theta} + \frac{p_\psi^2}{2 I_3} + M g \ell \cos \theta$$

The motion in θ is equivalent to a 1D particle in this effective potential.

8.4 Types of Top Motion

The θ -motion oscillates between two turning points θ_1 and θ_2 (where $E = V_{\text{eff}}(\theta)$). This oscillation in θ is called **nutation**. Meanwhile, ϕ changes, giving **precession**. The three qualitative behaviors are:

Nutation Patterns

1. **Looping:** $\dot{\phi}$ does not change sign — the tip traces loops (cusps or loops that go backward).
2. **Cuspidal:** $\dot{\phi} = 0$ at one turning point — the tip makes cusps on the sphere.
3. **Wavy:** $\dot{\phi}$ changes sign — the tip traces a wavy pattern between the turning circles.

8.5 Steady Precession

A special solution has $\theta = \theta_0 = \text{const}$ (no nutation). Setting $\frac{dV_{\text{eff}}}{d\theta}|_{\theta_0} = 0$ gives the **steady precession condition**. For a fast-spinning top ($I_3\omega_3 \gg \text{other terms}$):

Steady Precession Rate

$$\dot{\phi} \approx \frac{Mg\ell}{I_3\omega_3} = \frac{Mg\ell}{L_3}$$

The precession is slow and independent of θ_0 . Faster spin $\rightarrow$ slower precession.

8.6 Stability of the Sleeping Top

A top spinning vertically ($\theta = 0$) is called a **sleeping top**. It is stable (remains upright) provided the spin is fast enough:

$$\omega_3^2 > \frac{4Mg\ell I_1}{I_3^2}$$

Below this threshold, the top nutates and precesses.

9 Hamiltonian Mechanics

9.1 The Legendre Transform

Hamiltonian mechanics replaces the variables $(q_i, \dot{q}_i)$ with (q_i, p_i) via a Legendre transform.

The Hamiltonian

$$H(q_i, p_i, t) = \sum_i p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $\dot{q}_i$ is expressed in terms of q_i and p_i by inverting $p_i = \partial L / \partial \dot{q}_i$.

9.2 Hamilton's Equations of Motion

Hamilton's Canonical Equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

These are $2d$ first-order ODEs, equivalent to the d second-order E-L equations. They have a beautiful symmetry: q and p enter almost symmetrically (up to a sign).

Lagrangian vs. Hamiltonian

	Lagrangian	Hamiltonian
Variables	$(q_i, \dot{q}_i)$	(q_i, p_i)
Function	$L = T - V$	$H = \sum p_i \dot{q}_i - L$
Equations	d second-order ODEs	$2d$ first-order ODEs
Space	Configuration space $\mathcal{Q}$	Phase space Γ
Symmetry	Coordinate invariance	Canonical transformations

9.3 Phase Space

The $2d$ -dimensional space of $(q_1, \dots, q_d, p_1, \dots, p_d)$ is called **phase space**. Each point in phase space uniquely determines the future and past evolution of the system. Trajectories in phase space **never cross** (by uniqueness of solutions to ODEs).

9.4 Integrability

Definition 9.1 (Integrable System). *A Hamiltonian system with d degrees of freedom is (**Liouville**) **integrable** if it possesses d independent conserved quantities $I_1 = H, I_2, \dots, I_d$ that are **in involution** (their Poisson brackets all vanish pairwise).*

An integrable system can, in principle, be solved by quadratures (successive integrations). The heavy symmetric top with its three conserved quantities (E, p_ϕ, p_ψ) is a classic example.

10 Poisson Brackets

Poisson Bracket

For two functions $f(q_i, p_i, t)$ and $g(q_i, p_i, t)$ on phase space:

$$\{f, g\} = \sum_{i=1}^d \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

Fundamental Poisson brackets:

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}$$

The time evolution of any observable f is:

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}$$

In particular, f is conserved if and only if $\{f, H\} = 0$ (assuming no explicit time dependence). This gives a purely algebraic test for conservation laws.

11 Key Mathematical Tools

11.1 Calculus of Variations

The central mathematical technique underlying Lagrangian mechanics is the **calculus of variations**: finding functions that extremize functionals.

For a functional $J[y] = \int_a^b F(y, y', x) dx$, the extremal function satisfies:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0 \quad (\text{Euler-Lagrange equation})$$

This is the mathematical backbone of Hamilton's principle.

11.2 Euler's Homogeneous Function Theorem

If $f(\lambda x_1, \dots, \lambda x_n) = \lambda^k f(x_1, \dots, x_n)$ (homogeneous of degree k), then:

$$\sum_i x_i \frac{\partial f}{\partial x_i} = k f$$

This is used to prove that $H = T + V$ when T is quadratic in velocities (degree 2):

$$\sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} = 2T \quad \implies \quad H = \sum p_i \dot{q}_i - L = 2T - (T - V) = T + V$$

11.3 Levi-Civita Symbol and Cross Products

The **Levi-Civita symbol** ϵ_{ijk} is essential for angular momentum:

$$(\mathbf{A} \times \mathbf{B})_i = \sum_{j,k} \epsilon_{ijk} A_j B_k$$

Key identity: $\sum_k \epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$.

This appears frequently in the derivation of Euler's equations and the inertia tensor transformation.

11.4 Rotation Matrices

The rotation matrix $\mathbf{R}(\hat{\mathbf{n}}, \alpha)$ for rotation by angle α about axis $\hat{\mathbf{n}}$ satisfies: $\mathbf{R}^T = \mathbf{R}^{-1}$ and $\det \mathbf{R} = 1$.

An infinitesimal rotation $\delta \mathbf{r} = \delta \boldsymbol{\alpha} \times \mathbf{r}$ connects to the angular velocity:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

The full Euler-angle rotation matrix is the product of three successive rotations — this is what gives the angular velocity components in the body frame (Section 7.2).

12 Summary: The Analytical Mechanics Toolkit

The Big Picture

Analytical mechanics reformulates classical mechanics around a single scalar function (the Lagrangian or Hamiltonian) rather than vector forces. The payoff is enormous:

Lagrangian mechanics gives equations of motion that are automatically consistent with constraints, work in any coordinate system, and make symmetries manifest.

Noether's theorem turns every continuous symmetry into a conservation law — and vice versa.

Hamiltonian mechanics reformulates the problem in phase space, opening the door to statistical mechanics, chaos theory, and quantum mechanics (where Poisson brackets become commutators: $\{q, p\} = 1 \rightarrow [\hat{q}, \hat{p}] = i\hbar$).

Master Equation Sheet

Euler–Lagrange:	$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$
Conjugate momentum:	$p_i = \frac{\partial L}{\partial \dot{q}_i}$
Hamiltonian:	$H = \sum_i p_i \dot{q}_i - L$
Hamilton's equations:	$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$
Noether charge:	$I = \sum_i p_i K_i - F$
Euler's equations:	$I_i \dot{\omega}_i - (I_j - I_k) \omega_j \omega_k = \tau_i$
Poisson bracket:	$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$